

Monitoring of anaesthetised patient

The purpose of monitoring is to minimise the detrimental effects of excessively light levels of anaesthesia (i.e., awareness, recall, pain and movement) as well as those of excessively deep levels of anaesthesia (hypoventilation and hypoxaemia, reduced cardiac output, hypotension, inadequate tissue perfusion, hypothermia and prolonged recovery).

Monitoring should really start from preoperative patient assessment and should continue right the way through recovery.

Monitoring a patient physiological status provides us with warning signals when things go wrong, and allows early intervention and averted trouble.

Monitoring involves making measurements which are then compared to standard reference 'normal' values or ranges of values.

Aims of monitoring:

To maintain the function of certain body organ systems as close to physiological normality as possible.

To maintain adequate depth of anaesthesia (not too deep or light) under different levels of surgical stimulation during a procedure.

To ensure safety of the patient. Anaesthesia can carry a high risk for the patient. Horses especially suffer a high risk of morbidity and mortality under general anaesthesia (complication/death rate of 1 in 100 (all cases) to 1 in 200 (healthy); compared to 1 in 900 for cats, and 1 in 1800 for dogs (healthy animals)).

To ensure safety of personnel, maintain unresponsiveness of the patient to surgical stimuli.

Legal implications; Record events during an anaesthetic.

What should be monitored:

- 1) CNS
- 2) Cardiovascular system
- 3) Respiratory system
- 4) Temperature
- 5) Neuromuscular system
- 6) Renal system

1) Monitoring CNS:

Traditionally anaesthetists have utilised various reflexes of the CNS to help them assess the level of CNS responsiveness and from that, judge the depth of anaesthesia.

a) Assessment of reflexes:

- **Palpebral reflexes:** Check both eyes if possible, becomes refractory with over stimulation; affected if irritants such as hair, blood or surgical scrub enters eyes.
- **Corneal reflexes:** In most species it is present until deep/surgical levels of anaesthesia (although some times persists in horses). Regular touching of the cornea, whilst tear production is depressed, can damage the cornea.
- **Gag reflex and jaw tone:** Laryngeal and pharyngeal reflexes may persist until deeper planes of anaesthesia in pigs and cats than other species.
- **Limb withdrawal reflexes**
- **Perineal reflex (anal tone reflex)**
- **Righting reflex:** Often used for exotics ✓
- **Assessment of eye position, movement, lacrimation:**

Horses tend to maintain a slow palpebral reflex under greater 'depths' of anaesthesia than do dogs.

The eye ball position in horses, however, can be unreliable and often one eye has a different position from the other which is confusing.

Eye ball position in dogs, cats and ruminants is much more reliable. In dogs and cats eyes tend to rotate ventromedially (forward and downwards) as anaesthesia deepens towards the surgical plane, but the eyes become central if anaesthesia deepens further.

b) Assessment of autonomic nervous system activity:

The assessment of autonomic nervous system can also be used to help us gauge anaesthetic depth. If an animal is 'too light' then sympathetic nervous system is stimulated by painful surgery and catecholamines are released. These are responsible for increase in heart rate/ or blood pressure that we can observe. Small animals and ruminants tend to show both heart rate and blood pressure increase, whereas horses tend to show only increase in blood pressure with out very obvious change in heart rate.

* However occasionally if animals are too light and painful surgical stimuli occurs, there is fall in blood pressure and heart rate. This occurs because the animal has fainted. Remember to look for other signs of anaesthetic depth, such as eye reflexes and jaw tone, to help prevent you being confused.

c) Electroencephalography:

A more sophisticated way of monitoring CNS is EEG. EEG is the recording of the electrical activity of the cortical areas of brain, using the nearest location for electrodes (i.e., on the scalp)

The highest cost and difficulty of interpretation preclude this form of monitoring from routine veterinary use at the moment.

2) Monitoring cardiovascular system(CVS):

Preoperative assessment of CVS function is important. In order to ensure good patient survival, tissue oxygen delivery is the key player. As oxygen delivery to the tissues depends primarily upon the cardiac output and the oxygen content of arterial blood, ideally these two should be monitored.

mean arterial pressure - central venous pressure

Cardiac output = $\frac{\text{mean arterial pressure} - \text{central venous pressure}}{\text{systemic vascular resistance}}$

Cardiac output = stroke volume X Heart rate

Cardiac output measurement in veterinary science is mainly a research tool at the moment. The equipment necessary are usually very expensive. Examples are

- Thermodilution cardiac output
- Indicator dye dilution cardiac output
- Lithium dilution cardiac output
- Fick principle cardiac output
- TOE(trans oesophageal echocardiography)
- Bullet method

a)ECG: Essential for diagnosis of arrhythmias. The ECG cannot tell us if the heart is actually beating and, if so how well it is pumping blood around body, for example pulseless electrical activity(PEA) can occur where ECG looks completely normal but the myocardium does not contract.

Advantage of ECG is that , it allows electrical activity of heart and early detection of arrhythmias.

Disadvantage is that it does not give information on cardiac output, can be expensive and require adequate knowledge of cardiology.

b)Echocardiography:

In the form of TOE(trans oesophageal echo) is being used increasingly in human anaesthesia to image the heart's performance in the real time and even to measure cardiac output on a beat-to-beat basis.

c)Pulse:

This assess pulse rate, rhythm and quality / character.

Pulse pressure :The difference between the systolic and diastolic pressure. A pulse of 120/80 may feel similar to a pulse of 80/40 but the mean pressure would be different.

Mean pressure: We can estimate mean pressure by how easy it is to occlude the arterial pulse by digital pressure and by assessing how 'full' or 'turgid ' the artery feels between the pulses.

Character of pulse, for example we may describe a pulse as 'bounding' when a large difference between systolic and diastolic pressure exists. This is common in conditions such as aortic valvular insufficiency.

d) CVP

CVP gives a measure of the ability of the heart to cope with the volume of blood being returned to it. It indicates the balance between cardiac output and venous return.

The normal CVP in small animals is 0-10 cm H₂O. It is 15-30 cm H₂O in laterally recumbent horses, and 5-10 cm H₂O in dorsal recumbent horses.

Low and below low range values indicate hypovolemia and suggest that rapid bolus of fluid should be administered. Above range values indicate relative hypervolemia and that fluid therapy should be stopped.

CVP measurements are used to determine whether there is 'room' for additional fluid therapy in the management of hypotension.

e) Urine output:

This helps us determine the adequacy of circulating blood volume and peripheal tissues (including kidney) perfusion, but also tells us about renal functional/ dysfunction. Normal urine output is 1-1.5ml/kg/h

f) Mucous membrane colour and CRT:

This gives indication of state of circulation and peripheral perfusions, but remember that a dead animal can also give a normal CRT (≤ 2 s) because of backflow of blood from venous and arterial side. A CRT >4 s is significant. To perform CRT properly, blanch the mucous membrane for a good 5s.

g) Hematological analysis:

Occasionally it may be necessary to track changes in PCV, Hb and total protein during long surgical procedures, for example where fluid therapy support is needed.

h) Biochemical analysis:

During long surgical procedures it is useful to be able to track changes in electrolytes or glucose or urea and creatinine.

3) Monitoring of respiratory system:

We monitor respiratory system by observing

- a) Breathing rate
- b) Breathing rhythm

c) Tidal volume/depth of each breath

d) Mucous membrane colour

Minute ventilation = Breathing rate $\times$ tidal volume

Minute ventilation is also called as 'minute respiratory volume'

a) Breathing rate can be monitored by:

- Observe or palpate chest wall
- Observe movements of reservoir/rebreathing bag
- We can see movement of valves or hear 'chicking' of valves
- Esophageal stethoscopy
- Breathing rate often displayed by capnograph monitor.

c) Tidal volume:

It can be measured by wright's respirometer, spirometer or pneumotachometers.

d) Mucous membrane colour:

- This gives information about tissue perfusion
- White mucous membrane is associated with anemia.
- Yellow—Jaundice
 - Horses with lot of carotene in diet, the eyes look yellow.
- Grey/purple /cyanotic-associated with low tissue perfusion.
- Cyanosis is generally observed until there is atleast 5g/dl deoxyhemoglobin present.
- Navy blue mucous membrane - associated with excess N_2O administration.
- Cherry red mucous membrane – associated with carbon monoxide poisoning (eg large amount of carboxyhemoglobin).
- Muddy brown – associated with methemoglobin seen in nitrate poisoning, paracetamol poisoning, and prilocaine local anaesthetic technique.

Pulse oxymetry: To determine the degree of hemoglobin with oxygen saturation.

Gas and agent monitoring:

Routinely used by medical anaesthetists to verify that the oxygen is delivered through common gas outlet of anaesthetic machine.

4) Monitoring Neuromuscular System:

To ensure adequate neuromuscular blockage when and needed(for surgery).

To ensure adequate return of neuromuscular function when 'paralysis' no longer required.

- Check for Spontaneous respiration after thoracic surgery.

5) Temperature monitoring:

Some degree of hypothermia is common in many of our anaesthetized patients, and can be of causes of a prolonged recovery.

Temperature can be helpful in assessing the CNS of animal (eg Cold extremities , core periphery temperature gradients)

Anaesthesia generally depresses the 'thermostat' in the hypothalamus of the CNS and there by reduces a patient's ability to thermoregulate. Anaesthesia also prevents from shivering. Many drugs we use during anaesthesia cause peripheral vasodilation and thereby enhances heat loss. This is increased by clipping large patches of fur off (reducing insulation) and using copious volumes of scrub solution and then surgical spirit (wetting the animal and enhancing evaporative cooling). Preventing heat loss is easier than rewarming animal.

Hyperthermia can also be a problem. When we use rebreathing systems, soda lime canister produces heat and water while absorbing CO_2 (exothermic reaction) and the lights in the operation field also produces heat and when the theatre is not air conditioned then the surgeon may feel heat and at that time the animal may be hyperthermic.

6) Monitoring Renal system:

As many anaesthetics cause hypotension, and some may impair/alter renal function, it is important to consider the kidneys. For this urine output can be measured. Normal urine output is 1-1.5ml/kg/h. This requires bladder catheterization and initial emptying, so that we get a 'start' point.

Samples can be taken for dipstick analysis and in particular specific gravity, it can be affected by a number of factors during anaesthesia and plays important role in ICU.

The stress response to anaesthesia includes an increase in ADH secretion, and we can expect reduction in urine output during and for sometime after general anaesthesia. Some drugs (eg Alpha2 agonists) can effect urine output and whether patient receives i/v fluids during anaesthesia can effect and also influence urine output.

Often we administer i/v fluids to counter the fall in BP and try to support the circulation in already hypovolaemic animal is a must.